BARNABE **LEDOUX**
PhD student at ESPCI |

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PhD candidate in physics at the Gulliver Laboratory in ESPCI, with David Lacoste, on the topic “Growth in uncertain environments”. My curriculum includes courses in statistical physics, biophysics and physics of complex systems. I am particularly keen to integrate environmental considerations into my training, and hope to reconcile my research with these concerns during my PhD. I work at the frontier between physics and biology, but also economics.

Education

May 2026 **Beg Rohu, Saint-Pierre-Quiberon, France**

I followed the spring school Patterning principles in biophysics.
- **Physics and Biology** : Spatial patterns, Partial differential equations, Morphogenesis.

April 2025 **INTP, Surba, France**

I followed the spring school Mathematical Theory in Community Ecology.
- **Physics and Ecology** : Population dynamics, Coexistence, Diversity, Random matrices.

October 2024 **Nordita, Stockholm, Sweden**

I followed the program Measuring and Manipulating Non-Equilibrium Systems.
- **Physics** : Out of equilibrium statistical physics, Random walks, Entropy.

2023 - 2024 **École Normale Supérieure, Paris, France**

As a student in the ICFP (International Centre for Fundamental Physics and its interfaces) Master’s program, I followed the “Soft matter and biological physics” curriculum.
- **Physics** : Statistical physics (non-equilibrium systems), Statistical physics of complex and disordered systems, Ecology and evolution, Active matter and collective phenomena, Physics of biological systems, Physics of fluids and non-linear systems, Physics of interfaces.

2020 - 2024 **École Polytechnique, Palaiseau, France**

Polytechnique engineering student. I specialize in Physics, but have additional training in mathematics and biology. **Ranking 25/435** among École Polytechnique students.
- **Physics** : Statistical physics of condensed matter (electrons in solids, entanglement, mesoscopic quantum physics), Quantum physics, Physics of biological systems, Physics of living systems, Optoelectronics, Statistical physics, Relativity, Nonlinear optics, Digital physics, Electromagnetism.
- **Mathematics** : Analysis, Statistics, Linear Algebra.
- **Informatics** : Java, Python, Freefem ++.
- **Biology** : Molecular biology, Cell biology.
- **Humanities** : Literature, Philosophy, History of arts.

2018 - 2020 **Lycée Michelet, Vanves, France**

MPSI-MP preparatory class
- **Physics** : Mechanics, Optics, Electronics, Electromagnetism, Statistical physics, Quantum physics.
- **Mathematics** : Analysis, Algebra, Statistics.
- **Informatics** : Python.

2018 **Lycée Michelet, Vanves, France**

Baccalauréat mention très bien.

Experience

Today
September 2024

PhD, Laboratoire Gulliver ESPCI, Paris
PhD student in physics with David Lacoste on the topic "Population growth in an uncertain environment. This subject allows us to tackle questions of statistical physics, but also interdisciplinary projects at the frontier with ecology and biology.
Python

Today
Janvier 2026

Tutorial teacher in probabilities and statistics, ESPCI, Paris
- Mathematical basis for statistical physics.
- Correcting Jupyter Notebooks.

Today
Janvier 2025

Tutorial teacher in physics and chemistry for bachelor students, Lycée Henri IV, Paris
- Support students in their understanding of physics.
- Correcting homeworks.

Today
Janvier 2025

Scientific partner in primary schools, Fondation La Main à la Pâte, Paris
- Opening perspectives for students in priority education areas.
- Helping primary schools teachers to teach sciences.

July 2024
April 2024

Research internship, Laboratoire Gulliver ESPCI, Paris
- Working in the field of statistical physics applied to biological systems.
- Working with experimentalists.

Python

Juillet 2025

Official Marker IPHO (International Physics Olympiads), Société Française de Physique, Paris
- Correcting exams from the best physics students internationally.
- Debating with representatives from different countries.

September 2023
April 2023

Research internship, Cavendish Laboratory, Cambridge
- Working on areas at the cutting edge of condensed matter physics research. In particular, to study radiative effects in heat transport within solids using a quantum description and preparing conferences.

Python | Phonopy | Phono3py

September 2022
June 2022

Internship, RATP, Paris
- Data analysis of train wheel measurements.
- Physical study of a wheel defect.

VBA | Excel | Office

July 2024
September 2021

Evaluator in Physics-Chemistry, in MP and MPSI preparatory classes, Lycée Michelet, Vanves
- Evaluate students on physics-chemistry problems 3 hours per week.

Today
September 2021

Private teacher in physics-chemistry and mathematics, high school to university level,
- Help to understand and manipulate physical tools.

May 2021
August 2020

Internship, in the association "La Main à La Pâte", Le Havre
- Teaching and promoting science in primary school classes.
- Adapting complex scientific knowledge for pupils aged 9 to 12 (light, climate change, sound waves, electricity, robotics).
- Working with students from REP+ (priority education network).

Data science skills

Programming skills	**Python**, **VBA** (Virtual Basics for Application), Freefem++, Java.
Office	LaTeX, Pack Office (Word, Excel, PowerPoint).
Operating Systems	Windows , Linux , Android.

Works & Publications

2026 - ESPCI	**“Growth bistability in small bacterial populations exposed to antibiotics”** B. Ledoux, D. Lacoste **BiorXiv**
2026 - ESPCI	**“Compositional memory matters for early molecular systems”** B. Ledoux, R. Kuwabara, N. Ichihashi, R. Mizuuchi, D. Lacoste **PNAS** Vol. 123 | No. 16
2026 - ESPCI	**“Debt, Growth, and the Carbon Lock-In”** S. Montagnani, B. Ledoux, D. Lacoste **ArXiv**:2512.05063
2025 - ESPCI	**“Compositional memory matters for early molecular systems”** presented at the **Paris Biological Physics Community Day** and **Physics & Life 2025**.
2025 - ESPCI	**“Inhibition of bacterial growth by antibiotics”** B. Ledoux and D. Lacoste 2025 **Phys. Biol.** 22 066007
2025 - ESPCI	**“Universal features of autocatalytic growth”** part of **“Economic principles of cell biology”**
2025 - ESPCI	**“Limits to Growth in Uncertain Environments”** presented at the Clifrium 2025, Banque de France in Paris
2025 - ESPCI	**“Co-evolution in transient compartmentalization dynamics”** presented at the Gulliver-Centuri days in Marseille
2025 - ESPCI	**“Inhibition of bacterial growth by antibiotics”** presented at the Journées de Physique Statistique 2024 in Paris
2023 - Cavendish Laboratory	**Prize for research internship** in physics awarded by the École Polytechnique foundation.
2023 - Cavendish Laboratory	**“First-principles Wigner formulation of coupled radiative and conductive heat transfer”**. The project was presented at conferences.
2022 - École Polytechnique	Collective Scientific Project Report : Heat Perception . **Nominated among the 14 best joint scientific projects (out of 111 projects proposed in 2022)**.

Languages

- Score **TOEFL** : 107/120

French	●●●●●
English	●●●●●
German	●●○○○
Japanese	●●○○○

Forces

- Versatile
- Curious
- Loves to learn
- Hardworking
- Autonomous

Centres d’intérêts

Music :	Guitar (16 years), Bass guitar, Member of the school orchestra.
Literature :	English & American Literature.
Sports :	Climbing, Running, Hiking.